Customer Churn Analysis Report

Project Objective:

The objective of this project was to analyze customer churn behavior in a telecom company and develop machine learning models capable of predicting customers likely to discontinue services. The project focused on identifying key churn drivers, customer risk segmentation, and actionable business recommendations.

Data Preprocessing and Analysis:

The Telco Customer Churn dataset was cleaned and preprocessed using techniques such as handling missing values, encoding categorical variables, and feature engineering. Exploratory Data Analysis (EDA) and correlation analysis were performed to identify patterns related to customer churn.

Machine Learning Models

Three classification models were trained and evaluated:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

Among these models, Gradient Boosting achieved the best performance with the highest ROC-AUC score of 0.8377, making it the most effective model for churn prediction in this project.

Key Findings

- Customers with Month-to-Month contracts showed the highest churn rate compared to customers with long-term contracts.
- Higher MonthlyCharges and lower tenure were strongly associated with customer churn.
- Feature importance analysis identified TotalCharges, Tenure, and MonthlyCharges as the most influential churn-driving factors.
- High Risk customers were mostly customers with short tenure periods, high monthly charges, and no long-term contractual commitment.
- Risk segmentation showed that the majority of customers belonged to the Low Risk category, while a smaller portion required immediate retention attention.

Business Recommendations:

1. Provide targeted retention offers and loyalty incentives to customers identified as High Risk, especially during their first few months of subscription.

2. Encourage customers to move from Month-to-Month contracts to One-Year or Two-Year plans through discounts, bundled services, and reward programs.

Project Limitations:

The dataset did not contain customer satisfaction metrics, complaint history, internet usage patterns, or competitor pricing information, which may limit prediction accuracy. Additionally, class imbalance affected churn detection recall.

Future Improvements:

Future improvements could include applying SMOTE for class balancing, adding customer behavioral features, periodic model retraining, and implementing real-time churn prediction systems.

Conclusion:

This project successfully demonstrated how machine learning and customer analytics techniques can be used to predict churn behavior, identify high-risk customers, and support data-driven business decisions for improving customer retention.